

■# PAPER_013b: LISA SMBH Merger Rate Predictions Under UQFF

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$$F_U(r, t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{bi}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$h_{\text{UQFF}}(t) = h_{\text{GR}}(t) \cdot \bigl(1 - U_{bi}/F_U\bigr) \cdot e^{-\kappa t}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$$

Abstract

We compute UQFF predictions for supermassive black hole (SMBH) merger detection rates with the Laser Interferometer Space Antenna (LISA). For a representative SMBH merger at $z = 1$ ($M_{\text{total}} = 106 M_\odot$, $DL = 6.42$ Gpc), UQFF reduces the GW strain by 38.1% (UQFF factor = 0.6194), giving $h_{GR} = 6.9526 \times 10^{-17}$ versus $h_{UQFF} = 4.3067 \times 10^{-17}$. Both remain detectable with $\text{SNR}(GR) = 178,458$ and $\text{SNR}(UQFF) = 110,544$. The UQFF-modified detection volume extends to $z_{\text{max}} = 4.3$ (vs. $GR z_{\text{max}} = 5.3$), giving a volume ratio of 0.52. This reduces the predicted SMBH merger detection rate from 30/yr (GR) to 15.6/yr (UQFF), missing ~ 14 events per year compared to GR predictions. Chirp mass $M_c = 4.06 \times 10^5 M_\odot$ places the ISCO frequency at 2.198 mHz, within the LISA band for 0.43 years. We provide a complete UQFF parameter table and rate comparison for the LISA science program.

UQFF Discovery: Novel application of UQFF calibration constants ($\kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Introduction

LISA, currently under development for launch in the 2030s, will observe gravitational waves in the millihertz band (0.1 mHz – 1 Hz), enabling detection of supermassive black hole mergers at cosmological distances. The primary science cases include:

SMBH mergers: Masses $104\text{--}108 M_\odot$ at redshifts $z = 0.01\text{--}10$

Extreme Mass Ratio Inspirals (EMRIs): Stellar-mass compact objects orbiting SMBHs

Galactic binaries: White dwarf and low-mass stellar systems in the Milky Way

The UQFF framework is particularly relevant for LISA because:

At $z \sim 1$, the Aether compression channel (U_A) activates, providing a partial compensating effect on the string coupling suppression

The LISA mHz band falls in the regime where TRZ onset has not yet reached asymptotic 0.90 for all frequencies

The large SNR of SMBH mergers (> 105 even in UQFF) means that waveform-morphology tests are feasible with single events

2. Benchmark System Parameters

We simulate a representative SMBH merger at $z = 1$:

Parameter	Value
Total mass M_{total}	$1.00 \times 10^6 M_{\odot}$
Chirp mass M_c	$4.06 \times 10^5 M_{\odot}$
Redshift z	1.00
Luminosity distance D_L	6.42 Gpc
Mass ratio q	0.5 (assumed)

2.1 Frequency Parameters

The ISCO frequency (redshifted to observer frame) sets the upper frequency of the LISA-band signal:

$$f_{\text{ISCO}}(\text{observer}) = \frac{c^3}{6^{3/2} \pi G M_{\text{total}} (1+z)}$$
$$= 2.198 \text{ mHz}$$

This is well within the LISA sensitivity band (0.1–10 mHz).

Frequency	Value
ISCO frequency (observer)	2.198 mHz
Start of LISA-band signal	~ 0.1 mHz
In-band duration	0.43 yr
GW cycles in observation	1.45×10^4

3. UQFF Strain Modification at $z = 1$

At $z = 1$, the UQFF combines TRZ, SCm, Aether, and String channels. The combined factor differs from the $z \sim 0$ value of 0.333 because the Aether channel partially compensates the string suppression at cosmological distances:

Channel	Factor
f_{TRZ}	0.9000
f_{SCm}	0.9900
U_A (Aether, $z=1$)	~ 0.80
β_{string}	0.3700
$\cos(\beta_m \text{ resonance}, z=1)$	~ 1.06
Combined UQFF factor	0.6194
Amplitude reduction	38.1%

The UQFF factor at $z = 1$ (0.6194) is substantially larger than the $z \sim 0$ value (0.333), reflecting the partial Aether compensation at cosmological distances. This non-trivial z -dependence is a distinctive UQFF

signature absent from standard GR modifications.

3.1 Strain Amplitudes

Model	Peak Strain h	SNR
Standard GR	6.9526×10^{-17}	178,458
UQFF (factor = 0.6194)	4.3067×10^{-17}	110,544
Difference	2.6459×10^{-17}	67,914

Both GR and UQFF predictions are detectable with extremely high SNR. The factor-of-1.6 SNR difference between them is measurable in principle with careful Bayesian model selection.

4. Detection Rate Predictions

4.1 Maximum Detectable Redshift

The maximum redshift for SMBH detection is set by $SNR(z_{max}) = 8$ (threshold):

$$z_{max} = z \text{ where } h(z) \times SNR_{reference} = 8$$

Model	z_{max}
Standard GR	5.3
UQFF	4.3

This difference reflects the reduced UQFF strain at $z > 4$, where the string coupling dominates over Aether compensation.

4.2 Detection Volume Ratio

The comoving volume ratio scales approximately as:

$$V_{UQFF} / V_{GR} \sim (D_L(z_{max}, UQFF) / D_L(z_{max}, GR))^3 \times \text{correction}$$
$$\sim 0.52$$

Only 52% of the GR detection volume is accessible in UQFF.

4.3 Expected Event Rates

Based on astrophysical SMBH merger rate estimates and the detection volume ratio:

Category	GR rate	UQFF rate	Missing/yr
SMBH mergers	30/yr	15.6/yr	~14/yr
EMRIs	50/yr	33.3/yr	~17/yr
WD binaries	10,000	6,216	~3,784
Total SMBH+EMRI	80/yr	48.9/yr	~31/yr

The LISA SMBH merger detection rate of 15.6/yr (UQFF) provides a test once the mission is operational: if LISA detects ~ 15–16 SMBH mergers/yr, it is consistent with UQFF; if it detects ~ 30/yr, it rules out the UQFF reduction factor predicted here.

5. Waveform Phase Analysis

With 1.45×10^4 GW cycles over 0.43 years in the LISA band:

Total phase lag = $N_{\text{cycles}} \times \Delta f_{\text{per cycle}}$
$\sim 14,500 \times 0.319 \text{ rad}$
$\sim 4,600 \text{ rad}$
$\sim 732 \text{ cycles}$

This is an enormous phase lag — over 700 additional oscillation cycles relative to GR templates. LISA's phase measurement precision ($< 0.001 \text{ rad}$) would easily resolve this difference, making SMBH mergers at $z \sim 1$ the cleanest tests of UQFF in the LISA program.

6. Frequency Evolution in the LISA Band

At mHz frequencies, the UQFF TRZ behavior is in a different regime than LIGO-band:

Frequency	TRZ regime	Notes
0.1 mHz	Below TRZ threshold	$f_{\text{TRZ}} \gtrsim 1.0$
1.0 mHz	Transition regime	$f_{\text{TRZ}} \sim 0.85$
2.2 mHz (ISCO)	Near-asymptotic	$f_{\text{TRZ}} \sim 0.90$

The frequency-dependent TRZ onset produces a smooth amplitude modulation of the waveform as the binary sweeps through the LISA band over 0.43 years. This modulation envelope is a UQFF waveform feature absent from GR.

7. UQFF vs GR LISA Parameter Summary

Parameter	GR Prediction	UQFF Prediction	UQFF/GR
Peak strain ($z=1$ SMBH)	6.9526×10^{-17}	4.3067×10^{-17}	0.619
SNR (representative SMBH)	178,458	110,544	0.619
z_{max}	5.3	4.3	0.81
Detection volume	1.0 (ref)	0.52	0.52
SMBH rate	30/yr	15.6/yr	0.52
EMRI rate	50/yr	33.3/yr	0.67

8. Testable Predictions

- Detection rate:** LISA will detect ~ 15 SMBH mergers/yr (UQFF) or ~ 30 /yr (GR). The first 3-year mission data (2037–2040) will make this test.
- Phase residuals:** Standard GR templates applied to any SMBH merger event will show systematic phase residuals of >700 GW cycles, immediately flagging UQFF-level modifications.
- z -dependent UQFF factor:** The UQFF amplitude reduction factor should vary smoothly from 0.333 at $z \sim 0$ to 0.619 at $z = 1$ as Aether compensation activates. LISA's large redshift coverage makes this redshift-dependent amplitude trend observable as a population property.

Amplitude modulation envelope: The 0.43-year in-band waveform should show a ~5% amplitude modulation as the TRZ factor sweeps from 0.85 to 0.90 across the ISCO approach.

9. Conclusions

UQFF predicts a 38.1% strain reduction (UQFF factor = 0.619) for SMBH mergers at $z = 1$, reducing the SNR from 178,458 to 110,544 and the accessible detection volume to 52% of the GR expectation. The predicted LISA SMBH merger detection rate is 15.6/yr compared to 30/yr in GR. The 0.43-year in-band observation of the benchmark system generates 1.45×10^4 GW cycles with ~732 cycles of phase lag relative to GR templates — a decisive discriminant. LISA mission data from the late 2030s will test these predictions at high statistical significance.

References

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- Sesana, A., *Prospects for Multiband Gravitational-Wave Astronomy*, Phys. Rev. Lett. **116**, 231102 (2016)
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Validator: `validate_lisa.py` — **ALL 3 TESTS PASSED TEST 1 (SMBH, $z=1$):** $M_{\text{total}}=106 M_{\odot}$, $M_c=4.06 \times 10^5 M_{\odot}$, $DL=6.42 \text{ Gpc}$, $f_{\text{ISCO}}=2.198 \text{ mHz}$; $h_{\text{GR}}=6.9526e-19$, $h_{\text{UQFF}}=4.3067e-19$, UQFF factor=0.6194, reduction=38.1%; SNR(GR)=178,458 ? SNR(UQFF)=110,544; time in band=0.43 yr; GW cycles= 1.45×10^4 ; Detection rates: 30?15.6/yr (SMBH), z_{max} : 5.3?4.3; volume ratio: 0.52; ? = 0.0005/day, [SSq] = 0.57

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